Robotic Kidney Transplant Beyond the Learning Curve: 8-Year Single-center Experience and Matched Comparison With Open Kidney Transplant

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OBJECTIVE To investigate the medium to long-term outcomes of robotic-assisted kidney transplantation (RAKT) and propensity-matched comparison with open kidney transplant (OKT).

MATERIALS AND METHODS We retrospectively reviewed 342 patients from database, who underwent RAKT and OKT from January 2015 to May 2022, at our center. Various demographic, intraoperative, and post-operative parameters were evaluated. Statistical analysis including propensity matching using nearest neighbor algorithm was performed to ensure comparability between the RAKT and OKT groups. The statistical analyses were performed using SPSS v.22.0 (IBM Corp, Armonk, NY) and STATA 13 (Stata Corp, College Station, TX). All statistical tests were two-sided, and a significance level of $P < .05$ was considered statistically significant.

RESULTS After applying exclusion criteria, 196 RAKT patients and 102 OKT patients were included in the analysis. Propensity score matching resulted in the inclusion of 173 patients in the RAKT group. In the propensity-matched comparison of intra/perioperative parameters, RAKT showed significant reductions in total surgical time ($P < .001$), wound length ($P < .001$), blood loss ($P < .001$), blood transfusion rate ($P < .001$), pain score ($P < .001$), and analgesia requirement ($P < .001$). Graft survival and patient survival rates were comparable in RAKT and OKT groups at the end of 60 months.

CONCLUSION RAKT offers several advantages over OKT in terms of reduced operative time, blood loss, pain, and analgesia requirements. RAKT shows comparable graft and patient survival rates to OKT in the medium to long term. UROLOGY 183: 100–105, 2024.

Kidney transplantation is the gold standard treatment for end-stage renal disease, offering improved quality of life and long-term survival for patients.[1] Over the years, advancements in surgical techniques have led to the emergence of robotic-assisted kidney transplantation (RAKT) as a promising alternative to traditional open kidney transplantation (OKT).[2] The first recorded instance of RAKT took place in 2010, involving a morbidly obese patient.[3-5] Since then RAKT has been implemented in limited centers across the world.[5-8] Early studies primarily focused on evaluating the feasibility and noninferiority of RAKT when compared to OKT.[9-11] These pioneering studies have addressed the surgical outcomes, short-term graft function and complications.[12] It has been observed in initial series that RAKT is associated with longer re-warming and anastomotic times, yet the graft function remains comparable to that of OKT.[4] To date, there have been two studies that have specifically investigated the long and mid-term outcomes of RAKT. In a study spanning a period of 10 years and encompassing 239 patients, Tzveranov and colleagues examined the outcome of RAKT in morbidly obese patients, revealing minimal surgical site infections and excellent graft function.[13] Another study conducted by Ahlawat et al compared 126 RAKT patients with 378 patients who underwent open surgery.[14] This study indicated that RAKT was associated with lower blood loss, smaller incisions, reduced postoperative pain, decreased wound complications, and fewer symptomatic lymphoceles.[14] Our study focus on medium to long-term out comes of RAKT and propensity-matched comparison with OKT.

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The aim of our study is to investigate the medium to long-term outcomes of RAKT and provide a comprehensive comparison with OKT.

MATERIALS AND METHODS

We retrospectively reviewed 342 patients from prospectively maintained database, who underwent RAKT and OKT from January 2015 to May 2022, at our center, by single surgeon. The study was approved by the local ethics committee. The initial analysis included 212 patients who underwent RAKT. For matched comparison, pediatric patients less than 20 kg, deceased donor renal transplant, patients who had an open conversion from robotic were excluded. For OKT, we followed a standard technique previously described in literature using a modified Gibson incision. Our RAKT technique of total extra peritonealization of graft has been described in our previous publications.[8] All cases were performed by a single surgeon who had performed 606 OKTs and 1317 robotic procedures at the time of data analysis. Demographic variables included were, age, sex, body mass index (BMI), Charlson Comorbidity Index, history of previous transplant, presence of focal segmental glomerulosclerosis on kidney biopsy, duration of hemodialysis before kidney transplant, induction agent of the recipient and glomerular filtration rate (GFR), number vessels of graft kidney, human leukocyte antigen (HLA) A, B, DR match, presence of ABO incompatibility and side of nephrectomy of the donor (Table 1).

Intraoperative outcomes assessed were

Warm ischemia time - Duration between clamping the graft renal artery and placing the graft kidney onto an ice-slush bath.

Total ischemia time - Duration between donor kidney clamping and reperfusion of graft.

Rewarming time - Duration between placing graft kidney into the abdominal cavity and reperfusing the kidney following vascular anastomosis.

Anastomotic time - From the start of venous to the completion of arterial anastomosis.

Total operative time

Blood loss

Blood transfusion rate

Postoperative pain score was measured by Visual Analog Scale (0-5). The analgesia requirement, was calculated in injectable fentanyl requirement. Postoperative complications, graft rejection, and graft survival were assessed with a minimum 12 months follow-up. Delayed renal function was defined as, any patient having decreased urine output with requirement of hemodialysis in the immediate postoperative period. The maintenance treatment consisted of prednisone, tacrolimus and mycophenolate mofetil. Functional outcomes were assessed as postoperative nadir eGFR, time to nadir serum creatinine and periodic follow-up of eGFR by (MDRD formula)[15] at 1 week, 1 month, 3 months, 1 year, 3 years, 5 years, latest.

Statistical Analysis

Frequency distribution and simple percentages were used to describe categorical variables. Continuous variables were expressed as either the mean ± standard deviation or the median. To compare the distribution of continuous variables between groups, the Mann-Whitney test was utilized. The Pearson's chi-square test was employed to compare the distribution of categorical variables between groups. For survival analysis of graft and patient outcomes; Kaplan-Meier curves were generated. The two groups (RAKT and OKT) were compared for graft and patient survival using appropriate statistical methods. The statistical analyses were performed using SPSS v.22.0 (IBM Corp, Armonk, NY) and STATA 13 (Stata Corp, College Station, TX). All statistical tests were two-sided, and a significance level of $P < .05$ was considered statistically significant.

Propensity Score Analysis

To address potential confounding factors and ensure comparability between the RAKT and OKT groups, propensity score analysis was conducted. This analysis involved the use of eight preoperative clinical covariates as independent variables: age, sex, body mass index, duration of dialysis before transplant, Charlson comorbidity index, estimated glomerular filtration rate (eGFR), number of arteries in the graft kidney, and HLA matching. RAKT vs OKT was the binary dependent variable in the multiple logistic regression analysis. By performing the propensity score analysis, we aimed to eliminate potential confounders that could influence the outcomes and to create well-matched groups. A 1:1 match was carried out between the RAKT and OKT groups using the nearest neighbor algorithm. The matching process ensured that patients in the two groups had similar characteristics for any significant differences identified in the covariates. This matching strategy resulted in the inclusion of 173 patients in the RAKT group, with RAKT serving as the control group for the comparison between the two cohorts.[16]

RESULTS

During the study period, a total of 342 kidney transplants were performed, including 212 cases of RAKT and 126 cases of OKT (Table 1). After applying the exclusion criteria, 196 patients in the RAKT group and 102 patients in the OKT group were compared (Table 1). The propensity-matching strategy resulted in the inclusion of 173 patients in the RAKT group, with RAKT serving as the control group for the comparison between the two cohorts.

Propensity-Matched Comparison of Intra/Perioperative Parameters

There was a significant reduction in favor for RAKT total surgical time ($P < .001$), wound length ($P < .001$), blood loss ($P < .001$), blood transfusion rate ($P < .001$), pain score from day 1-4 ($P < .001$) and corrected analgesia requirement ($P < .001$). The anastomotic time ($P = .752$), rewarming time ($P = .9$), total ischemia time ($P = .85$), delayed graft function ($P = .585$) and hospital stay were similar in both groups ($P = .026$) (Table 2).

Propensity Matched Comparison of Postop Parameters

Medical complications ($P < .001$) and graft loss ($P < .001$) was more in OKT group. Conversely, there were no significant difference in re-exploration rate ($P = .499$), wound infection ($P = .283$), incisional hernia ($P = 1.0$), lymphocele formation ($P = .371$), and urinary leak. Medical complications ($P < .001$) and graft loss ($P < .001$) was more in OKT group. Incidence of delayed graft function ($P = .485$), immunological events ($P = .044$), and death ($P = .341$) were similar in both groups (Table 3). The e-GFR was comparable in both cohorts at different time point up to 5 years (Supplementary Fig. 1).

Table 1. Association between group and demographic profile.

Parameters	Group			After Propensity Analysis		
	Open (n = 102)	RAKT (n = 196)	*P* Value	OKT (173)	RAKT (173)	*P* Value
Recipient factors						
Age (y)	37.66 ± 13.14	40.22 ± 11.63	.099[1]	39.38 ± 12.75	40.08 ± 11.30	.911[1]
Male (%)	78 (76.5%)	148 (75.5%)		128 (74.0%)	129 (74.6%)	.902[2]
BMI (kg/m²)***	21.73 ± 4.24	23.52 ± 4.26	.001[1]	23.67 ± 4.75	23.62 ± 4.39	.918[3]
Duration of HD before transplant (d)	170.26 ± 209.17	214.40 ± 224.59	.098[3]	218.49 ± 217.90	214.40 ± 224.59	.544[1]
History of previous transplant (yes)	3 (2.9%)	11 (5.6%)	.395[4]	1 (0.6%)	11 (6.4%)	.003[2]
Charlson Comorbidity Index	2.66 ± 1.01	2.62 ± 1.03	.699[3]	2.71 ± 1.09	2.59 ± 0.99	.411[1]
FSGS (yes)	8 (7.8%)	12 (6.1%)	.573[2]	11 (6.4%)	11 (6.4%)	1.000[2]
Induction agent						
Basiliximab (yes)	44 (43.1%)	93 (47.4%)	.479[2]	94 (54.3%)	83 (48.0%)	.237[2]
ATG (yes)	6 (5.9%)	18 (9.2%)	.320[2]	13 (7.5%)	15 (8.7%)	.693[2]
Other induction (yes)	0 (0.0%)	4 (2.0%)	.303[4]	0 (0.0%)	4 (2.3%)	.123[4]
No induction (yes)	52 (51.0%)	82 (41.8%)	.132[2]	67 (38.7%)	71 (41.0%)	.661[2]
Single vessel (1 RA, 1 RV)	70 (68.6%)	146 (74.5%)	.282[2]	131 (75.7%)	126 (72.8%)	.539[2]
Total months of follow-up ***	55.3 ± 22.9	27.6 ± 20.3	< .001[3]	52.9 ± 27.5	25.5 ± 20.1	< .001[1]
Donor factors						
Donor: age (y)	47.28 ± 9.97	47.37 ± 10.21	.943[1]	46.23 ± 8.90	46.92 ± 10.31	.552[1]
Female (%)	73 (71.6%)	152 (77.9%)	.223[2]	118 (68.2%)	132 (76.3%)	.093[2]
Donor: ABO incompatibility (yes)	0 (0.0%)	4 (2.0%)	.303[4]	0 (0.0%)	4 (2.3%)	.123[4]
Donor: HLA Match 0	16 (15.7%)	26 (13.3%)	.569[2]			
HLA Match 1–3	78 (76.5%)	151 (77.0%)	.912[2]	25 (14.5%)	23 (13.3%)	.756[2]
HLA Match 4–6	8 (7.8%)	20 (10.2%)	.507[2]	136 (78.6%)	132 (76.3%)	.607[2]
				12 (6.9%)	18 (10.4%)	.252[2]
Donor: DTPA left (mL/min)	46.47 ± 7.02	46.35 ± 7.24	.867[3]			
DTPA right (mL/min)	47.13 ± 7.48	47.61 ± 7.69	.639[3]	46.14 ± 6.95	46.21 ± 7.27	.934[1]
Donor: side of nephrectomy			.059[2]	47.70 ± 7.98	47.51 ± 7.86	.952[1]
Right	12 (11.8%)	11 (5.6%)				.645[2]
Left	90 (88.2%)	185 (94.4%)		9 (5.2%)	11 (6.4%)	
				164 (94.8%)	162 (93.6%)	

ATG, antithymocyte globulin; BMI, body mass index; DTPA, diethylenetriamine pentaacetate; FSGS, focal segmental glomerulosclerosis; HD, hemodialysis; HLA, human leukocyte antigen; OKT, open kidney transplantation; RAKT, robotic-assisted kidney transplantation.
*** Significant at $P < .05$. 1: *t*-test, 2: Chi-square test, 3: Wilcoxon-Mann-Whitney U test, 4: Fisher exact test.

Table 2. Intra and perioperative parameters (after propensity matching).

	OKT (173)	RAKT (173)	
Anastomotic time (min)	38.64 ± 7.48	38.13 ± 6.47	.752
Rewarming time (min)	40.62 ± 7.46	40.71 ± 6.47	.90
Total ischemia time (min)***	80.67 ± 16.69	80.36 ± 14.58	.85
Console time (min)***	NA	99.47 ± 22.43	
Total surgical time (min)***	197.60 ± 64.69	175.08 ± 24.96	< .001[1]
Wound length (cm)***	14.89 ± 0.90	6.63 ± 0.99	< .001[1]
Blood loss***	96.53 ± 82.58	62.05 ± 53.60	< .001[1]
Blood transfusion***	0.28 ± 0.50	0.14 ± 0.52	< .001[1]
Delayed function (yes)	8 (4.6%)	6 (3.5%)	.585[2]
HD after transplant (yes)	2 (1.2%)	7 (4.0%)	.174[4]
Pain Score			
(VAS) (day 1)***	2.79 ± 1.22	1.72 ± 0.87	< .001[1]
(VAS) (day 2)***	1.98 ± 0.51	1.23 ± 0.58	< .001[1]
(VAS) (day 3)***	1.53 ± 0.80	0.62 ± 0.68	< .001[1]
Corrected analgesia requirement***	904.45 ± 554.37	470.77 ± 209.72	< .001[1]
Duration of hospital stay (d)***	7.39 ± 1.96	7.62 ± 1.88	.026[1]

*** Significant at $P < .05$, 1: Wilcoxon-Mann-Whitney U test, 2: Chi-square test, 3: *t*-test, 4: Fisher exact test.

Table 3. Comparison of perioperative and long-term complications between OKT and RAKT.

Complications	OKT (173)	RAKT (173)	
Re-exploration (yes)	3 (1.8%)	4 (2.4%)	.499^{4}
Wound dehiscence (yes)	6 (3.5%)	2 (1.2%)	.28^{4}
Wound infection (yes)	3 (0.0%)	3 (1.8%)	1.00^{4}
Incisional hernia	1 (0.57)	0 (0.0%)	1.00
Lymphocele (yes)	1 (0.57%)	4 (2.4%)	.37^{4}
Urinary leak (yes)	0 (0.0%)	2 (1.2%)	.49^{4}
Others surgical complication (yes)	4 (2.4%)	2 (1.2%)	.68^{4}
Surgical complication not related to transplant surgery (yes)	5 (2.9%)	8 (4.6%)	.39^{2}
Medical complication (yes)***	45 (26.0%)	13 (7.5%)	< .001^{2}
Immunological event/Rejection (yes)***	36 (20.8%)	22 (12.7%)	.044^{2}
Delayed function (yes)	8 (4.6%)	6 (3.5%)	.585^{2}
Graft loss (yes)***	28 (16.2%)	6 (3.6%)	< .001^{2}
Death (yes)	14 (8.4%)	10 (5.8%)	.341^{2}

*** Significant at $P < .05$, 1: Wilcoxon-Mann-Whitney U test, 2: Chi-square test, 3: *t*-test, 4: Fisher exact test.

Graft Survival in RAKT and OKT

There was no significant difference in death-censored graft function at the end of 60 months in OKT (93.7 months) and RAKT was (91.3 months) (Log-rank P = .235) (Supplementary Fig. 2).

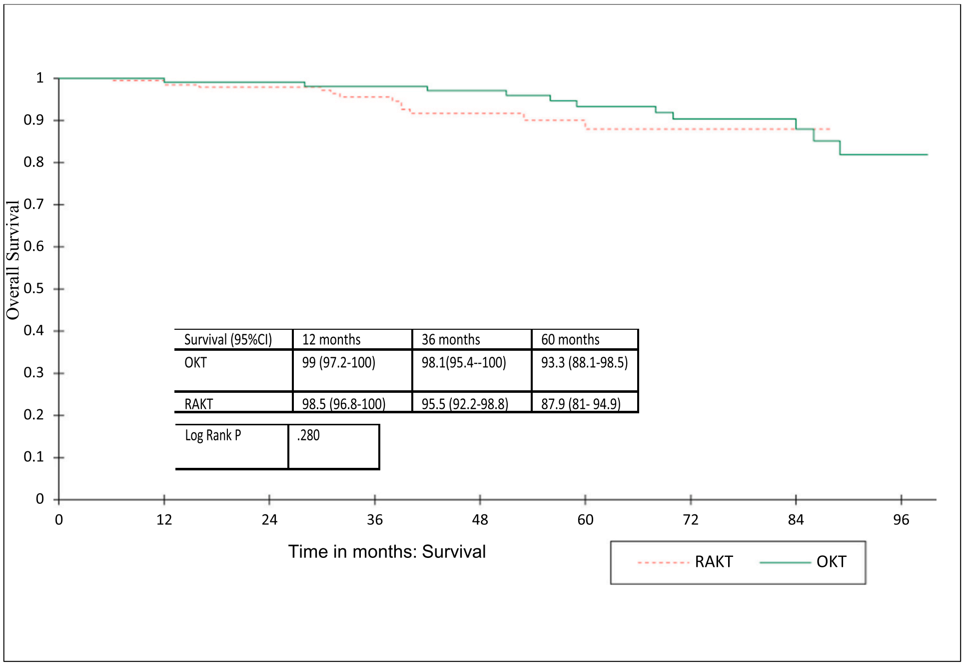
The patient survival at the end of 60 months in OKT 93.3 months and RAKT 87.9 also did not significantly differ between two groups (Log-rank P = .280) (Fig. 1).

RAKT Complications

In our RAKT series, we encountered six cases requiring an open conversion, two of which were due to arterial thrombosis. The first case was attributed to a technical fault while the second case was due to underlying hypercoagulability. Both the grafts were salvaged with immediate open conversion. Both these patients required two sessions of hemodialysis in the immediate postperiod and recovered subsequently. Four out of six patients had transient vascular spasm which improved upon open conversion. we encountered two cases of ureteric leaks, which was repaired through a Pfannensteil incision. In addition, we had two cases that required re-exploration within the first 24 hours due to hemorrhage attributed to surface bleeder and underlying coagulation defect.

DISCUSSION

RAKT was initiated over a decade ago with first published results focused on its utility among obese patients who had been denied OKT.[3] Since then, some centers have offered RAKT with patients with normal BMI also.[6] Various technical modifications have been proposed to enhance the outcomes and reduce risks.[8,12,17] Concerns were raised about the safety and feasibility of the procedure. In contrast to procedures like robotic radical prostatectomy, RAKT has not gained instantaneous acceptance worldwide. This lack of acceptance can be attributed to complex nature of the



Survival (95%CI)	12 months	36 months	60 months
OKT	99 (97.2-100)	98.1(95.4--100)	93.3 (88.1-98.5)
RAKT	98.5 (96.8-100)	95.5 (92.2-98.8)	87.9 (81- 94.9)

Log Rank P	.280

Figure 1. Kaplan Meier curve comparing the overall survival in OKT and RAKT patients. OKT, open kidney transplant; RAKT, robotic-assisted kidney transplantation. (Color version available online.)

surgery, limited robotic expertise among transplant surgeons and high stakes involved in the procedure. To date, the largest study published from single center is that of 239 patients conducted by Tzvetanov et al.[13] Other notable publications include multicentric European study of 320 patients and another large series of 126 patients by Ahlawat et al.[9,14] To our knowledge this is the second largest single-center series published thus far.

The primary concern surrounding RAKT in the initial era were longer operative time and rewarming time.[17] Significantly longer rewarming times have been reported in previously published literature.[4,5,18] To mitigate longer detrimental effects of warm ischemia time Sood et al proposed the regional hypothermia technique.[12] We have adopted a similar technique by introducing a GeLPOINT (Applied Medicals, CA) through Pfannensteil.[8] Cold ischemia is achieved by placing ice over the kidney while introducing kidney through GeLPOINT and subsequent reinforcement by placing ice while during anastomosis. However, as our proficiency increased requirement for additional ice has been minimized. Placement of large quantity of ice can hinder vascular anastomosis due to constant seepage of water into the field. Intensive use of intra-abdominal ice slush was considered as one of the contributing factors to the development of paralytic ileus.[7]

As we progressed beyond the learning curve in RAKT rewarming time has considerably decreased. Our RAKT rewarming time of 40 minutes is slightly less than what has been reported in previous studies, which varied from 47 to 73 minutes in various institutions.[6,7,13] Reduced rewarming time may be attributed to a larger proportion of normal BMI patients and improved surgeon experience in our series. Another noteworthy finding of our study was the significant reduction in operative time compared to OKT in RAKT cases. This reduction in operative time can be attributed to the efficiency of the RAKT procedure, specifically in terms of faster opening, bed preparation, and closure times. Challenging situations like multiple vessel, atherosclerotic vessel, right side allograft with short renal vein and second transplant can also be undertaken by robotic-assisted method. Currently, our contraindication are pediatric patients less than 20 kg and patients with previous multiple intraabdominal surgeries.

Concerns about pneumoperitoneum and its effects on graft kidney were initially raised. Studies have shown that at standard pneumoperitoneal pressures (12-15 mmHg), renal blood flow decreased to 25% of baseline, which becomes more relevant once the graft is reperfused.[19] To alleviate the effect of pneumoperitoneal insufflation we reduce the pressure to 8 mm of Hg after vascular anastomosis. In patients with normal BMI ureteric anastomosis can easily also be accomplished through the GeLPOINT placed in Pfannenstiel incision therein not exposing the reperfused graft to detrimental effects of pneumoperitoneum.

Reduced postoperative pain has been consistently observed in almost all the studies and our findings align with previous published literature. This can be attributed to smaller nonmuscle cutting Pfannensteil incision in RAKT. In contrast to previous published paper, we did not notice any difference in postoperative hospital stay between two cohorts. This may be influenced by the strict adherence to hospital protocol of discharging patients on postoperative day 6. While cost has often been cited as a drawback of RAKT, it is important to consider the specific setting and context of our study. In our setting, we found that the additional charges incurred for utilizing RAKT were only 2000 USD. This cannot be generalized to Western countries, where baseline expenses are typically higher, therefore the cost of RAKT may indeed be more substantial.

The benefit of reduced wound complications in RAKT has been described, particularly in studies including patients with high BMI. However, we did not observe any reduction in wound complications in the RAKT group, possibly due to a larger proportion of patients with normal BMI in our cohort. Reduced rate of lymphocele has been reported as advantage in selected studies.[5,6] Authors have hypothesized that RAKT, being an intraperitoneal procedure, prevents accumulation of lymph in the extraperitoneal space. However, we could not demonstrate any significant reduction in lymphocele formation. This could be attributed to our technique of total extraperitonealization, were the graft placed in extraperitoneal space.[8]

Vascular spasm has been our commonest cause of open conversion. Four out of six patients had transient spasm which improved upon open conversion. We have observed vascular spasm particularly in patients where there is excessive traction on the kidney during repositioning into the iliac fossa. To address this issue, we implemented a technique to perfuse the graft for a few minutes before repositioning. The use of Doppler scan and indocyanine green (ICG) has been instrumental in identifying cases of vascular spasm. In instances where vascular spasm occurs, it is recommended to reduce the Trendelenburg's position (head-down tilt) and lower the insufflation pressure. These maneuvers aim to alleviate the tension and traction on the blood vessels, minimizing the risk of vascular spasm and its associated complications.

While many studies have described a delay in achieving nadir creatinine, we did not observe any significant difference in our patient cohort.[4] This could be attributed to comparable rewarming times with increased experience.[4] There was no statistical difference between e-GFR at various time points of 1 week, 1 month, 3 months, 1 year, 3 years, and 5 years.

Our study possesses both strengths and limitations that are important to acknowledge. On the positive side, it represents the second-largest single-center series of RAKT and includes a substantial number of patients with a follow-up period ranging from 1 to 8 years. This extensive follow-up allows for a comprehensive evaluation of the long-term outcomes and provides valuable insights into the evolution of RAKT beyond the initial learning curve. Nevertheless, it is critical to recognize the limitations of our study. First, it is a single-surgeon study, which restricts the generalizability of our findings to a broader population. The outcomes observed in our study may be influenced by the specific surgical expertise and techniques of the participating surgeon. Replication of our results by multiple surgeons is necessary to validate the findings. Moreover, the retrospective nature of the analysis introduces inherent limitations. The absence of randomization and the potential for selection bias may impact the comparability of the RAKT and OKT groups. Although propensity matching was performed, we observed a significant difference in follow-up time between the two cohorts, which could have influenced the outcomes. The longer follow-up duration in the OKT group may have led to increased medical complications and graft loss in the OKT group.

Furthermore, certain patient groups were excluded from the analysis, including pediatric patients weighing less than 20 kg, deceased donor renal transplant recipients, and those who required conversion from robotic surgery to an open procedure. While these exclusions were made to ensure a fair and matched comparison, they may limit the generalizability of our findings to these specific patient populations.

CONCLUSION

Our study demonstrates that RAKT, once the learning curve is overcome, offers several advantages over OKT. RAKT shows a reduction in operative time, comparable rewarming time, and similar short to medium-term and long-term graft function when compared to OKT. One of the inherent benefits of RAKT is its minimally invasive nature, which leads to decreased postoperative pain and blood loss. These findings highlight the potential of RAKT as a viable alternative to OKT, offering improved surgical outcomes and patient experiences.

Declaration of Competing Interest

The authors have no conflict of interest to declare.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.urology.2023.10.031.

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